WORKED SOLUTION

STEP II 2014, Question 10

Mechanics • Envelope of Trajectories

STEP II 2014, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s2-q10

This question has a full guided treatment in *Beyond STEP*, Section 11.9, *The Safety Parabola*: https://beyondstep.keithmcnulty.org/ch11/safety-parabola.html

The key idea

Once we've removed the time, the trajectory equation is a *quadratic in* $\lambda = \tan\theta$, because $\sec^2\theta = 1 + \lambda^2$. Fix a point (x, y) and ask whether some launch angle passes through it. That's the same as asking whether this quadratic in λ has a real root, so the discriminant gives the boundary of the reachable region. For the final part, the boundary curve turns out to be a parabola with its focus at O, and that makes distances from O very easy to work out.

Solution

The trajectory

Model the projectile as a particle moving freely under gravity, with no air resistance. After time t,

$$x = ut\cos\theta, \qquad y = ut\sin\theta - \tfrac{1}{2}gt^2.$$

Since $\cos\theta \neq 0$ (the angle is less than a right angle), $t = x/(u\cos\theta)$, and substituting gives

$$y = x\tan\theta - \frac{gx^2}{2u^2\cos^2\theta}.$$

Using $\sec^2\theta = 1 + \tan^2\theta = 1 + \lambda^2$, the Cartesian equation is

$$\boxed{y = \lambda x - \frac{gx^2}{2u^2}\left(1 + \lambda^2\right).}$$

The maximum height Y above a given x

Fix u and fix $x > 0$. Now y is a function of λ alone, and we can rewrite it as

$$\frac{gx^2}{2u^2}\lambda^2 - x\lambda + \left(\frac{gx^2}{2u^2} + y\right) = 0.$$

Ask yourself Why think of this as an equation for λ rather than for y?

A value y occurs above this x exactly when some launch angle produces it, that is, when this quadratic has a real root λ. Its discriminant must be non-negative:

$$x^2 - 4 \cdot \frac{gx^2}{2u^2}\left(\frac{gx^2}{2u^2} + y\right) \geq 0.$$

Dividing by $x^2 > 0$ and then multiplying by $u^2/(2g) > 0$, which doesn't change the direction of the inequality,

$$\frac{u^2}{2g} \geq \frac{gx^2}{2u^2} + y \iff y \leq \frac{u^2}{2g} - \frac{gx^2}{2u^2}.$$

So

$$\boxed{Y = \frac{u^2}{2g} - \frac{gx^2}{2u^2}}$$

and no trajectory goes higher than Y above this x.

We also need every value *up to* Y to be reached, using a genuine launch angle with $\lambda > 0$. As a function of λ, y is a downward-opening quadratic. The vertex is at $\lambda_0 = u^2/(gx) > 0$, where $y = Y$ (substitute and check). For $\lambda \geq \lambda_0$, y is continuous and decreasing, and $y \to -\infty$ as $\lambda \to \infty$. So by continuity every value $y \leq Y$ is taken for some $\lambda \geq \lambda_0 > 0$, that is, for some $\theta \in (0, \frac{\pi}{2})$. Hence the values of y form the whole interval $y \leq Y$, as required. (At $x = 0$ we have to throw straight up. Then y goes from 0 to $u^2/(2g)$, which is again $Y(0)$.)

If we're allowed to project in either direction, the case $x < 0$ is the mirror image, and the same formula holds because Y depends only on x^2.

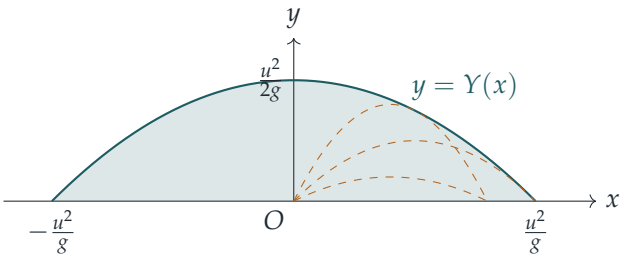
The graph and the reachable region

The graph of Y is an inverted parabola. Its vertex is at $(0, u^2/2g)$, the greatest possible height, and it meets the ground at $x = \pm u^2/g$, the greatest possible range. Every individual trajectory lies under this curve and touches it once, at $\lambda = \lambda_0$. For this reason it's often called the *bounding parabola* or *parabola of safety*.

The particle stops once it hits the ground. Take a point with $y \geq 0$ on a trajectory with $\lambda > 0$ and $x > 0$. That trajectory is above the ground for all x between O and the point, so the particle really does reach the point before landing. So the reachable set is the region

$$0 \leq y \leq \frac{u^2}{2g} - \frac{gx^2}{2u^2}, \qquad |x| \leq \frac{u^2}{g},$$

shaded below. The figure uses units of u^2/g, and the dashed curves are three sample trajectories.



The furthest distance from O

Any reachable point P lies on the segment from O to the point Q where the ray OP meets the boundary. The region is bounded by the bounding parabola and the ground, and it contains every point between O and its boundary along each ray. So $OP \leq OQ$, and we only need to maximise the distance over the boundary. On the ground the furthest points are at distance u^2/g. On the curved part, $y = Y(x)$, so

$$\begin{aligned} OQ^2 = x^2 + Y^2 &= x^2 + \left(\frac{u^2}{2g}\right)^2 - 2 \cdot \frac{u^2}{2g} \cdot \frac{gx^2}{2u^2} + \left(\frac{gx^2}{2u^2}\right)^2 \\ &= \left(\frac{u^2}{2g}\right)^2 + \frac{x^2}{2} + \left(\frac{gx^2}{2u^2}\right)^2 = \left(\frac{u^2}{2g} + \frac{gx^2}{2u^2}\right)^2. \end{aligned}$$

So $OQ = \dfrac{u^2}{2g} + \dfrac{gx^2}{2u^2} = \dfrac{u^2}{g} - Y$. In other words, the distance from O equals the distance down from the line $y = u^2/g$: O is the focus of the bounding parabola and $y = u^2/g$ is its directrix. This distance is largest when Y is smallest, and on the reachable part of the curve the least value is $Y = 0$, at $x = \pm u^2/g$. Hence the furthest distance from O that a projectile can reach is

$$\boxed{\frac{u^2}{g}},$$

which is the maximum horizontal range, achieved at $\theta = \frac{\pi}{4}$. Every point at a height $y > 0$ on the boundary is closer, by exactly y.

Checks

- Put $x = 0$: $Y = u^2/(2g)$, the height reached when throwing straight up (from $v^2 = u^2 - 2gh$). Put $Y = 0$: $x = u^2/g$, the familiar maximum range $u^2 \sin 2\theta/g$ at $\theta = \frac{\pi}{4}$.
- Take $u^2/g = 10$ (say $u = \sqrt{98}\ \mathrm{m\,s^{-1}}$ with $g = 9.8$). At $x = 4$ the formula gives $Y = 5 - 0.8 = 4.2$, reached at $\lambda_0 = 10/4 = 2.5$: indeed $2.5 \cdot 4 - \frac{1}{20} \cdot 16 \cdot 7.25 = 10 - 5.8 = 4.2$.

- A computer search over many launch angles and times, for $u = 10\ \mathrm{m\,s^{-1}}$, found the furthest point from O at distance $10.204\ \mathrm{m} = u^2/g$, on the ground.

Common slips

- Stopping at the discriminant. It shows that $y \leq Y$ is *necessary*, but the question also wants every such value to be *attained* with a genuine launch angle above the ground.
- Shading the whole region under $y = Y$, including points below the ground. The particle stops when it lands, so the reachable set also has $y \geq 0$.
- Assuming the furthest point is the top of the parabola of safety. That point is only $u^2/(2g)$ from O, half the range.

Beyond STEP

The bounding parabola is the *envelope* of the family of trajectories: the curve that touches every member of the family. One standard way to find an envelope is to eliminate the parameter between $F(x, y, \lambda) = 0$ and $\partial F/\partial\lambda = 0$, which is just the double-root condition we used. There is an energy link too: every trajectory has the same directrix, the line $y = u^2/(2g)$, which is the height at which all of the kinetic energy would have turned into potential energy. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers projectile motion and energy at first-year level.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.